

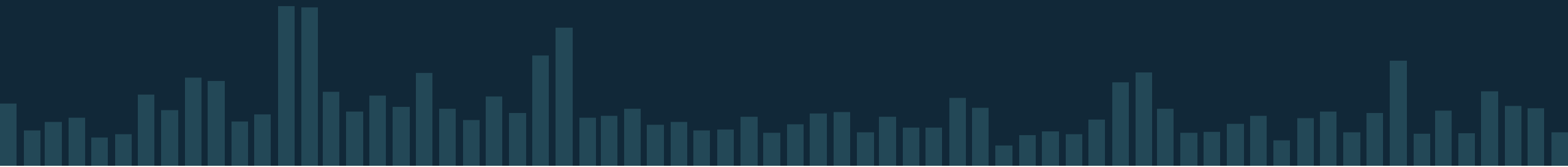
Reading the economy through the News: AI-assisted narrative and sentiment analysis of media text

| *How visible is the central bank when inflation fills the news?*

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Hrvatsko katoličko sveučilište

Based on the working paper by Petra Palić and Luka Sikić, version of 28 September 2026



Outline

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KEYWORDS

central bank communication

media attention

inflation

text as data

time series

AI-assisted text analysis

Croatia

One question, one measure, one hypothesis

TOPIC

How much room HNB, the Croatian central bank, gets in online news about inflation, 2021 to 2026

RESEARCH QUESTION

Does HNB's share of inflation news shrink when inflation rises?

AIM

Build a transparent, reproducible text measure of that share and test its link with inflation

HYPOTHESIS

Crowding out: higher inflation brings more inflation news in total, so HNB's slice gets smaller

What we build on, and the gap

Limited attention

Mankiw and Reis 2002 · Sims 2003

Inflation and attention

Malmendier and Nagel 2016 · Bracha and Tang 2019

News selection and framing

Entman 1993 · Shoemaker and Vos 2009

Central bank communication

Woodford 2003 · Coibion, Gorodnichenko and Weber 2022

Methods

Newey and West 1987 · Jordà 2005

RESEARCH GAP

We know a lot about whether central bank messages move expectations. We know little about how much of the inflation conversation the bank actually holds in the news, and how that share moves with inflation.

Data and methods

DATA

Croatian digital media archive

42.3 million news items and posts, 1 January 2021 to 10 September 2026, with headline, text, source, reach and tone label

Two text corpora

Items naming HNB or the central bank, and all inflation news

Inflation

Croatian HICP, annual rate, Eurostat

Checks

EC consumer survey, euro area food and energy HICP, FAO Food Price Index

METHODS

Measure

Keyword search, rule based prominence score, reach weighted share and attention gap

Model

Time series regression, 295 weeks and 67 months, Newey West errors

Robustness

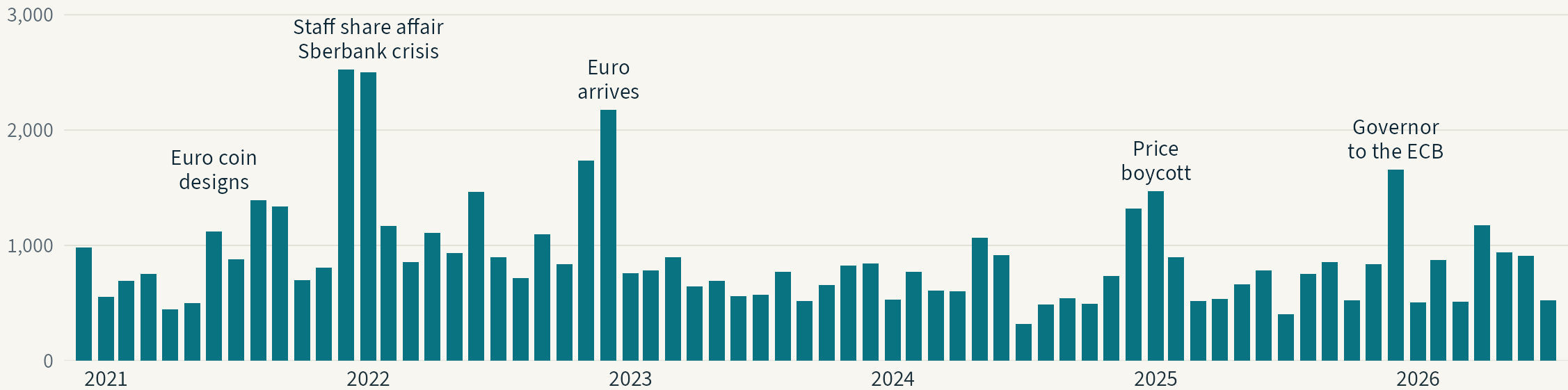
Alternative measures, leave one month out, block permutation, instruments

Dynamics

Local projections on consumer price expectations

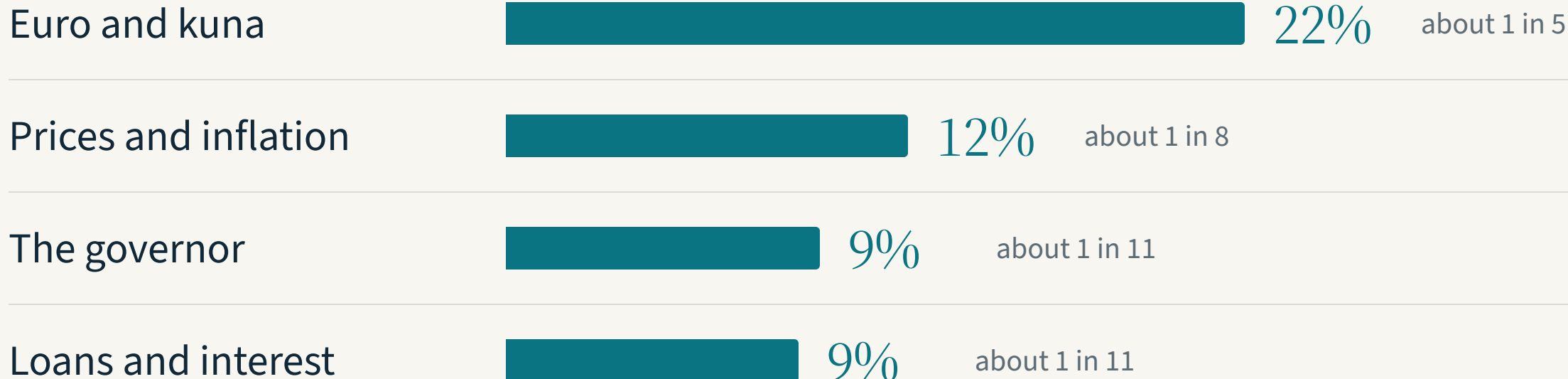
HNB is in the Croatian news every day

29 items a day name HNB
2,403 sources · January 2021 to August 2026



Euro and prices lead the headlines around HNB

Share of headlines that mention each topic, among items naming HNB



In 2022 HNB had more inflation stories but a smaller slice

	2021		2022	
Inflation stories with HNB	8,419	→	17,278	twice as many
All inflation stories	40,639	→	129,932	three times as many
HNB's slice	21%	→	13%	smaller

More stories, smaller slice.

A headline story counts more than a passing mention

OUR ARTICLE · ENGLISH SUMMARY

HNB expects average inflation of 7.7% in 2023

Index.hr · 30 June 2023

0.97 HNB at the centre

COMPARISON · ENGLISH SUMMARY

Minister demands a return to earlier prices

Index.hr · 5 January 2023

0.07 One passing mention

Score from 0 to 1,
four questions

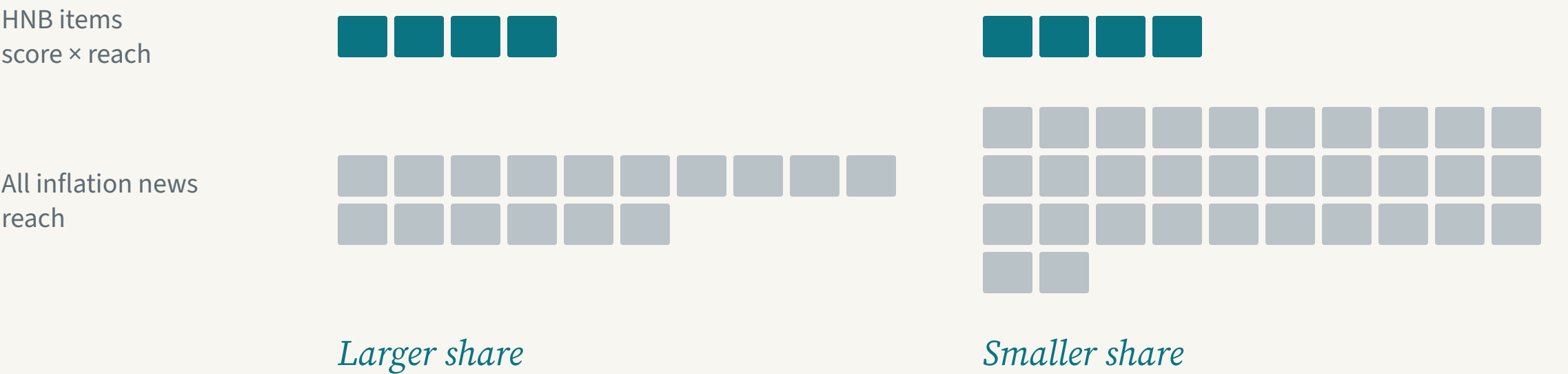
In the headline?

Named early?

Named often?

Acts or is quoted?

More inflation news shrinks HNB’s share



HNB SHARE

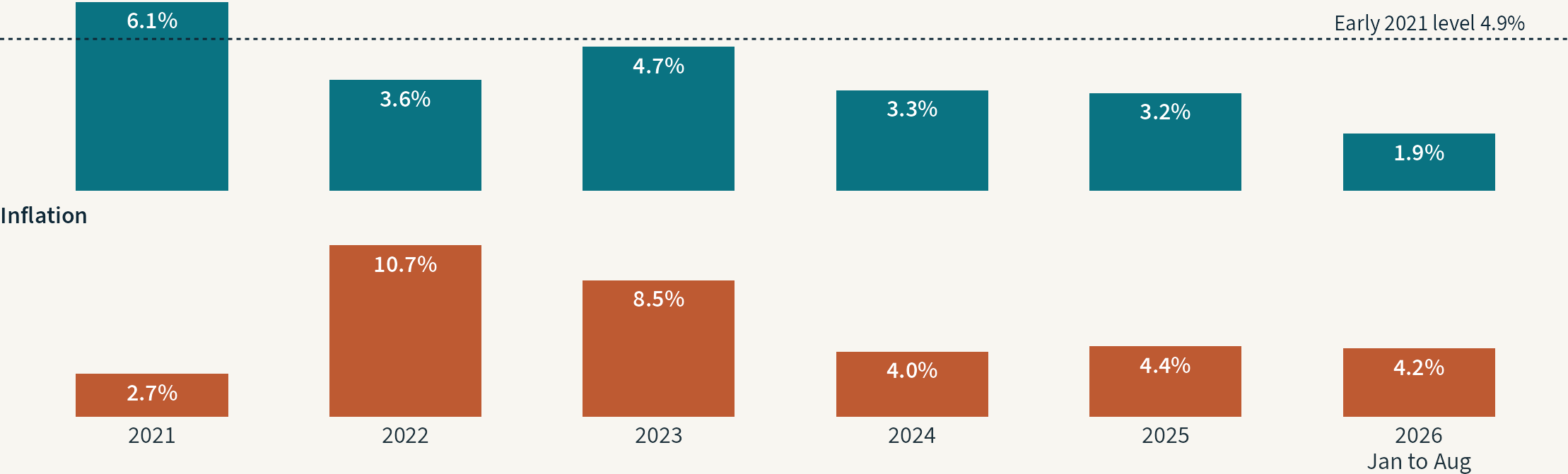
teal pile ÷ grey pile

ATTENTION GAP

early 2021 share, about 5%, minus today's share

HNB’s share fell in the inflation surge and stayed low

HNB share of inflation news



Our June 2023 article is one of 13,365 HNB inflation items that year.

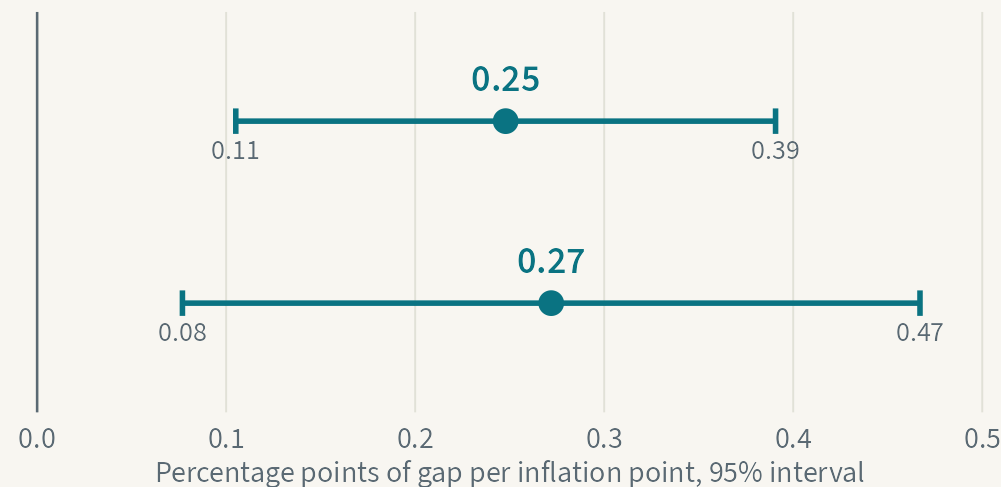
Higher inflation goes with a wider attention gap

0.25

percentage points wider gap
per point of inflation
for the same amount of HNB coverage

Weekly model
295 weeks

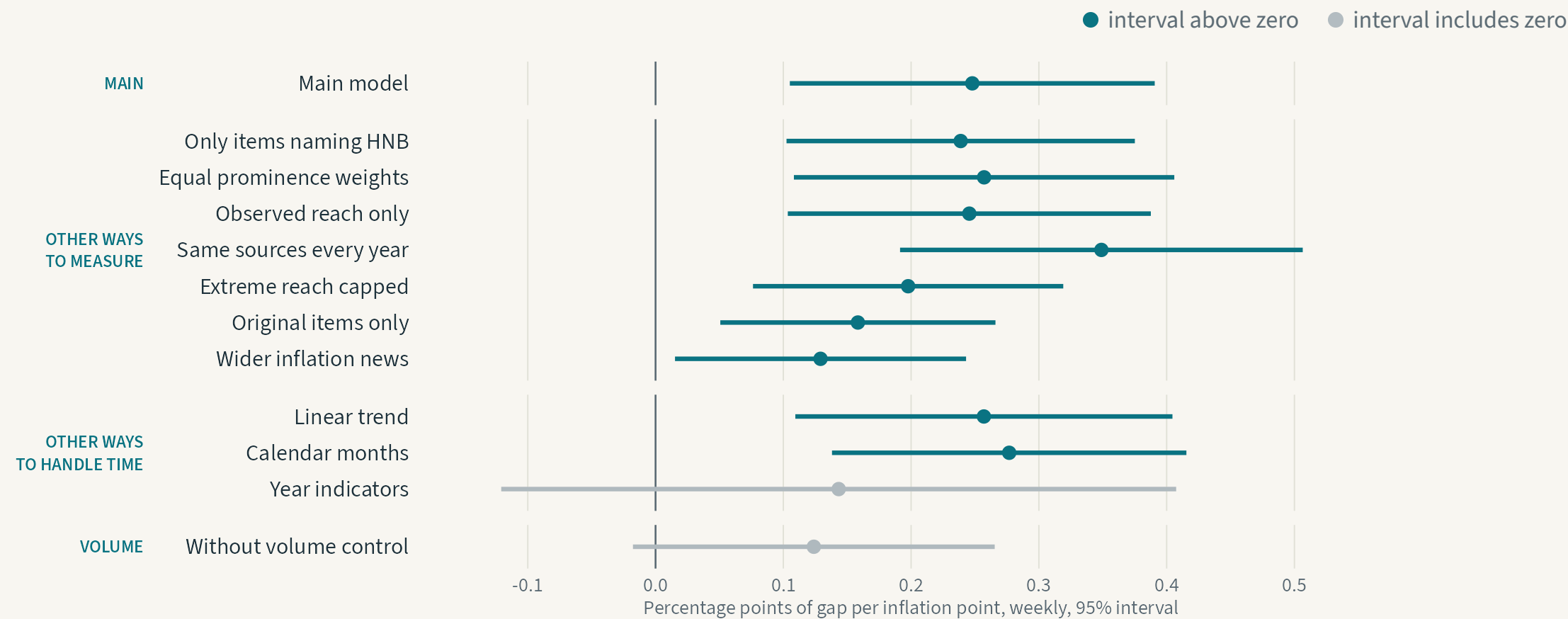
Monthly model
67 months



2% → 10% inflation ≈ gap about 2 points wider

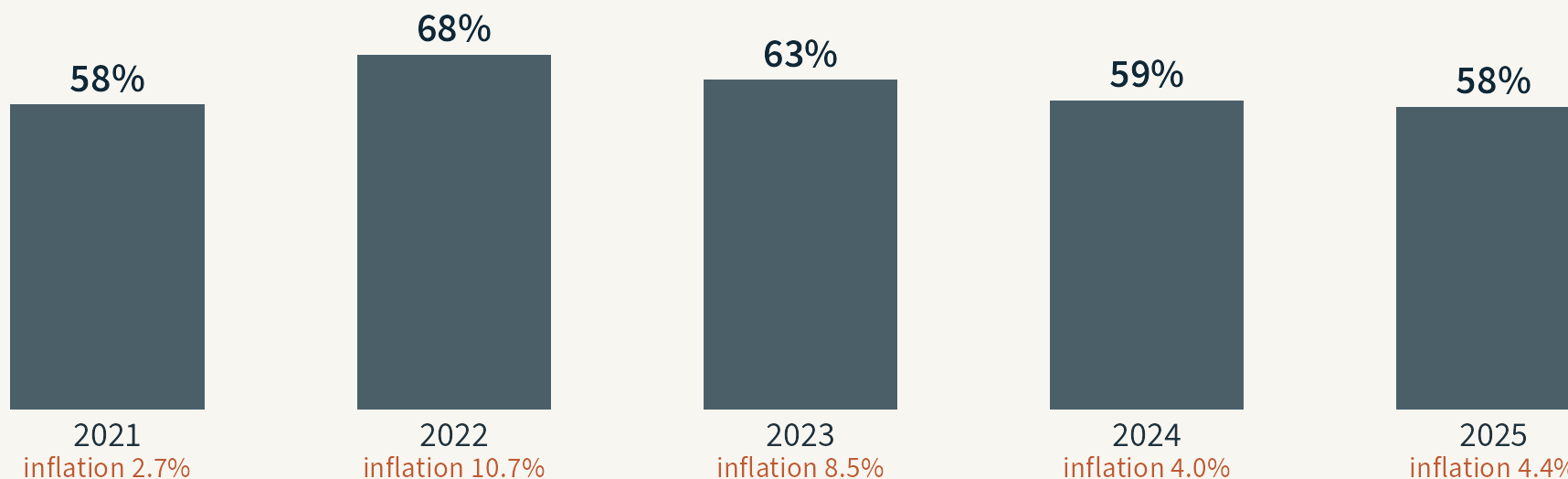
The starting share
was about 5%

Robust across measures, sensitive to year controls



Most HNB inflation news is labelled negative

Share of HNB inflation items with a negative tone label



CAVEAT The labels come from the data vendor and have not yet been checked against human readers

AI can check what our simple rules miss

PLANNED · NOT YET DONE

1

FIND

Which bank? Is it even news?

2

WEIGH

How central is HNB in the story?

3

TONE

Is the tone label right?

4

STORY

What explanation does the article give?

THE TEST

400 articles coded by two people, then the AI is checked against them

Our article shows what was said, not what was understood

OUR ARTICLE · ENGLISH SUMMARY

HNB expects average inflation of
7.7% in 2023

Index.hr · 30 June 2023 · score 0.97

- ✓ Published
- ✓ HNB at the centre
- ⊙ Read
- ⊙ Understood
- ⊙ Changed expectations

Conclusions and limitations

CONCLUSIONS

Crowding out

Higher inflation goes with a smaller HNB slice of inflation news, 0.25 weekly and 0.27 monthly

Count, not reach

The link runs mainly through fewer HNB stories relative to all inflation stories

No return

After inflation fell, HNB's share stayed low

Expectations

No stable link between the gap and later consumer price expectations

LIMITATIONS

Association

Not a causal effect, and sensitive to year indicators

Reach

A weight for possible audience, not a count of readers

Keywords

Central bank can mean the ECB, and tone labels are the vendor's

Text only

Shows what was said, not what was understood

Three key takeaways

- 1 HNB is in the news every day, but its slice of inflation news shrank when inflation rose, and has not come back
- 2 For the same HNB coverage, one point more inflation goes with a gap 0.25 points wider
- 3 AI can sharpen every step, if we check it against people

Next step: read the story behind the headline.

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